

IT2080 -Information Technology Project

Assignment 1 - Project Proposal



Smart Coconut Plantation Management System

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Background

Company Background

“Agro Plantation” is a coconut plantation company located in Sri Lanka. It focuses on blending technology with traditional farming to boost productivity and sustainability. With years of experience in growing, processing, and selling coconuts and their byproducts, the company is an important part of the island’s agricultural economy. Known for its quality practices and eco-friendly methods, CocoSmart operates across several hectares of fertile coconut farms in both coastal and dry regions of the island. The company has initiated a plan to digitize its operations by implementing a centralized management system to streamline labor, inventory, and sales.

What is Sri Lankan Coconut Farming?

Sri Lanka is one of the leading producers of coconuts. Coconuts are a crucial export and a key element of local diets. The country's coconut triangle, which includes the Kurunegala, Puttalam, and Gampaha districts, is the center of coconut farming. Coconut farming in Sri Lanka includes several stages, such as land preparation, planting, fertilizing, irrigation, pest control, harvesting, and post-harvest processing. Although traditional methods have been used for a long time, there is a growing effort to modernize farming practices with agricultural technology and smart farming techniques. However, small- and medium-sized plantations often struggle with issues like inefficient labor, unpredictable weather, poor record keeping, and limited use of digital tools, leading to a need for more efficient solutions.

What is a Coconut Plantation Management System?

A Coconut Plantation Management System is a digital tool designed to automate and improve the main operations of a coconut farm. This system includes features for labor management, task assignment, inventory management, sales processing, financial reporting, plantation tracking and even AI-based support for crop care. It allows different users such as HR managers, field supervisors, inventory officers, and sales managers to access real-time data and manage their tasks through one platform. The system may also connect with external tools like weather APIs for climate-related farming decisions and payment gateways for easy transactions. In the context of Sri Lanka’s coconut sector, this kind of centralized platform not only boosts operational efficiency but also promotes transparency, lowers costs, and enhances resource use.

Our Web Application

The Smart Coconut Plantation Management System, also known as CocoSmart, is a web-based platform designed to digitize and simplify daily operations on a coconut plantation. The system supports different user roles, including Admin, HR Manager, Field Manager, Inventory Manager, and Sales Manager. Each role has access to personalized dashboards and tools related to their tasks.

Through the platform, users can manage and monitor key activities. They can assign labor tasks, track plantation health, handle inventory, process product sales, and coordinate deliveries. CocoSmart also features an online marketplace, enabling the plantation to list and sell coconut products directly to customers. Users can securely manage payments and view sales data in real time. Furthermore, the system incorporates modern technologies like AI-based crop care assistance and weather-based decision support. This helps plantation staff make informed choices while improving productivity, accuracy, and overall farm efficiency.

By combining these features, CocoSmart aims to transform the coconut industry in Sri Lanka. It offers real time management, better worker coordination, transparent operations, and smarter decision making.

Problem and motivation

Problem Statement

Coconut plantation owners face increasing challenges in efficiently managing their operations due to the limitations of manual processes. From monitoring tree health to managing labor, stock, and sales, the lack of a centralized digital system leads to inefficiencies, delays, and revenue losses. This project aims to address these issues by developing a smart, web-based solution tailored to streamline and digitize the entire plantation management process.

Current Problem and Process

Our client runs a coconut plantation where almost all activities are handled manually. This includes recording tree details on paper, verbally assigning daily tasks to workers, managing stock in notebooks, and handling sales through local buyers or middlemen. There is no structured way to track performance, view real-time plantation health, or make data-based decisions. Due to this:

- Tree health data is rarely updated or monitored over time.
- Workers' tasks and schedules are poorly organized, leading to confusion or delays.
- Inventory records are incomplete or outdated, causing either overstocking or shortages.
- Sales opportunities are missed because the client depends only on wholesalers and has no system to directly reach customers.
- There's no unified dashboard to monitor the plantation's operations, analyze data, or plan ahead.

Overall, the manual process results in poor visibility of plantation performance, difficulty in managing workforce accountability, and limited market access, ultimately affecting growth and profitability.

Motivation

By addressing the above challenges through a centralized and intelligent web application, the client stands to benefit in multiple ways:

- **Improved Operational Efficiency:** With digital tracking of tree health, labor tasks, and inventory, the client can manage day-to-day operations with better accuracy and control.
- **Better Decision Making:** Access to real-time reports and insights will help the owner make informed, data-driven decisions on plantation care, sales trends, and workforce management.

- Expanded Market Reach: Introducing a customer-facing marketplace enables direct sales, helping the client move beyond just wholesaler-based selling, thus increasing profits.
- Stronger Workforce Accountability: A clear task management system ensures that each worker knows their responsibilities, and performance can be tracked effectively.
- Minimized Waste and Losses: Timely alerts on low-stock, expiring products, and upcoming tasks prevent unnecessary delays or spoilage.
- Scalability: With a digital system in place, the client can easily scale operations in the future without increasing administrative workload.

Ultimately, this system not only modernizes the client's existing plantation operations but also opens up new opportunities for growth, profitability, and long-term sustainability.

Aim and objectives

Aim

The main goal of the Coconut Plantation System is to create and design an intelligent, AI-powered Coconut Plantation System that automates plantation functions, helps new farmers, enables better decision-making, and improves overall productivity. The system integrates a variety of main modules such as field monitoring, inventory control, e-commerce, employee management, and user engagement into one easy, integrated platform.

Objectives

1. Develop a centralized plantation management module

This module will facilitate the easy maintenance and updating of plantation owners' information regarding their coconut fields and trees. Users will be able to schedule field activities like irrigation, fertilizer, and pest control and record them to analyze in the future. It will also remind them of pending activities, and allow them to monitor tree health manually or sensor-based.

Why it's relevant: It facilitates the management of farms and nothing goes unaddressed, which can directly affect productivity and crop quality.

2. Add an AI chatbot assistant

There will be an interactive chatbot to guide new or beginner farmers. It will answer questions, provide step-by-step instructions for coconut farming, diagnose tree ailments through keywords or images, and suggest which products are best sellers in the market.

Why it matters: It serves as a 24/7 farm helper, aiding users in making better decisions even when they are inexperienced.

3. Have an inventory management system in real-time

This module allows raw and processed stock such as coconuts, oil, packing material, and tools to be monitored. It possesses features to revise stock quantities, handle batches, and monitor expiry along with low-stock alerts. The process will be streamlined through QR/barcode scanning.

Why it matters: Having control over the inventory avoids waste and shortages, boosting operational efficiency.

4. Create a simple-to-use online market place

This e-commerce module allows owners to list their coconut produce (e.g., husk, oil, coconut water) with prices, pictures, and amounts in stock. Users can register, order, and pay online.

Why it matters: It generates new sources of income through direct-to-customer sales, enhancing profit margins and visibility.

5. Have a comprehensive workforce management module

Plantation owners can register workers, assign them to tasks like pruning or harvesting, and track their attendance and performance. The system can calculate working hours and corresponding payments.

Why it matters: It ensures fair pay and efficient task allocation, leading to enhanced labor productivity and cost savings.

6. Develop a robust user and role management system

This module supports managing various user roles such as admins, plantation workers, and consumers. System functionality is controlled by restricting access to roles, and users are allowed to give feedback or report issues. AI is applied in analyzing and categorizing feedback to identify themes or issues.

Why it matters: Proper access control ensures security, while feedback analysis allows ongoing improvement of the system based on user experience.

7. Offer actionable dashboards and reports

There will be an individual dashboard per module showing real-time information and graphical charts. Examples are plot health status, worker activity, stock levels, and earnings. The system will produce historical performance, sales, and feedback trends reports.

Why it's important: Visual data enables managers to immediately spot issues, monitor progress, and make data-driven decisions.

8. Improve operational efficiency and decision-making

By combining automation, data processing, and AI inputs, the system helps farmers make better choices. Market information can help determine the most suitable products to grow or sell.

Significance: It saves time, reduces wastage, and optimizes productivity and revenues of coconut plantation businesses.

System overview

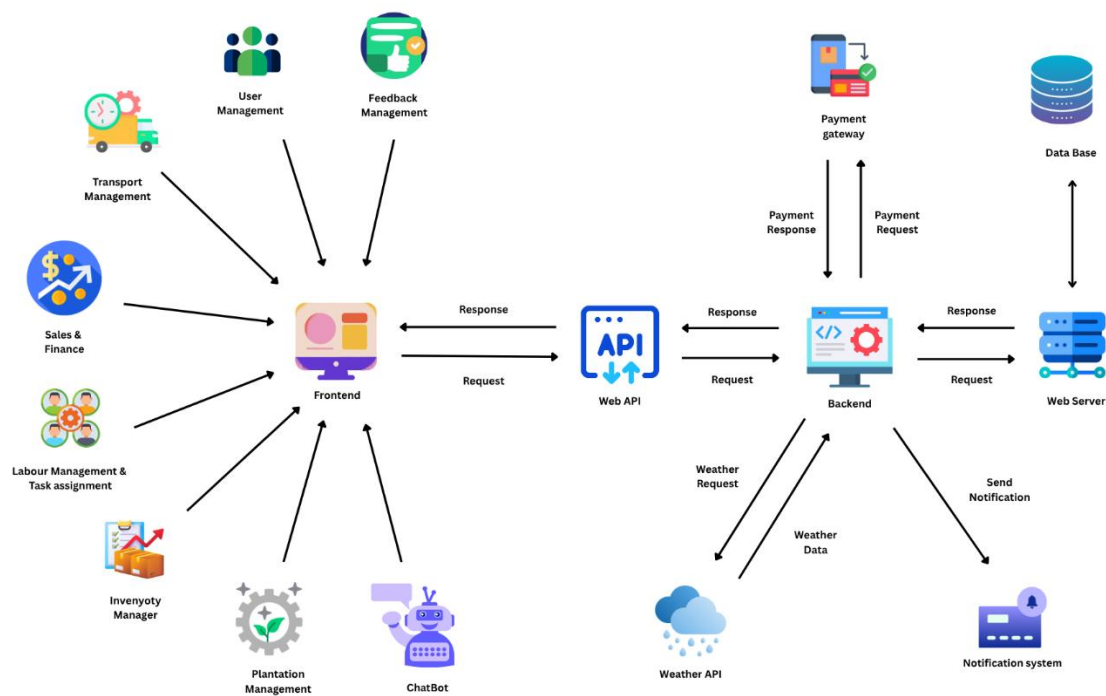


Figure 1

Plantation Management and Chatbot Module

This function is designed to simplify day-to-day plantation operations of coconuts and offer AI-powered field activities, and utilizes assistance to experienced and inexperienced farmers alike. It allows for a centralized plotting of plots, trees, and chatbot to offer real-time guidance and recommendations.

The users can register coconut plots and delineate boundaries for them. The trees can then be added, edited, or removed within these plots. Metadata such as age, type, planting date, and health status can be defined for every tree, allowing farmers to monitor and administrate their plantation over the internet.

Farmers can have complex schedules for irrigation, fertilizer, and pest control activities. This ensures regularity and avoids under or over-treatment.

The system can be designed to handle updates (e.g., entering viewed symptoms or visual observations). With this data, the system can notify which trees should be addressed.

The system will also automatically prompt reminders and notifications for overdue or pending tasks. These can be pushed via the dashboard, SMS, or mobile push notifications, so nothing vital field operation will get missed.

For helping new coconut farmers and reducing dependence on external experts, this module has a native chatbot that is able to:

- Answer Frequently Asked Questions (FAQ)
- Provide step-by-step guidance on how to grow coconuts
- Identify diseases based on symptom keywords or uploaded images
- Provide market demand trends for coconut products
- Support voice and text inputs for accessibility

The chatbot can be reached from both the main dashboard and the landing page, so assistance is always at hand.

Benefits:

- Digitizes and consolidates field management
- Optimizes resource usage through intelligent suggestions
- Supports new farmers with on-demand AI assistance
- Helps to increase production and reduce errors in operations

Inventory Management Module

The Inventory Management component of the Coconut Plantation Management System is designed to help plantation staff and managers efficiently track, manage, and control all stock-related activities, both for raw materials and finished coconut-based products. This module plays a key role in minimizing wastage, ensuring timely replenishment, and maintaining an organized and cost-effective inventory system.

Users with authorized access can add new inventory items, edit existing records, or delete products that are no longer in use. Each inventory item is clearly marked as either a product item (e.g., coconut oil, coir products) or a purchased item (e.g., fertilizers, packaging materials, tools), facilitating better classification and stock control.

The system enables users to input and monitor key details for each item, such as item code, name, description, unit price, reorder level, quantity, expiry date (for perishable items), and batch information. To avoid unexpected shortages, users can define reorder levels, and the system will automatically generate low-stock alerts when these thresholds are reached.

To improve organization, items can be assigned to specific storage units or categories, making it easier to locate and manage them. A search bar facilitates searching for a product/s by its item code. Additionally, the system supports QR code generation and scanning, allowing users to

quickly update or view stock information through a mobile device or scanner, ideal for real-time warehouse operations.

A central dashboard provides a clear overview of current stock levels, items approaching expiry, and critical alerts. This enables plantation staff to make quicker decisions and stay proactive in managing the inventory.

For better insights and planning, the system generates monthly reports covering stock inflow and outflow, product spoilage, and turnover rates. These reports help in identifying trends, reducing waste, and optimizing purchasing and production schedules.

To maintain security and accountability, only users with proper authorization can access or manage the inventory. The system also allows entering the purchase cost or production cost of each item, and automatically calculates the average batch cost, which is reflected in the inventory reports. This helps in evaluating profitability and planning resource allocation more accurately.

In summary, this Inventory Management module provides a reliable, organized, and cost-effective way to handle the plantation's stock, supporting the broader goal of improving productivity and minimizing operational losses.

User Management

CocoSmart will provide a secure and organized way for different types of users to interact with the system. During registration, each user will be able to choose their role—Admin, Worker, Processing Staff, or Customer—and the system will ensure that they only have access to the features relevant to their responsibilities.

Users will log in using a unique username and password, and security will be enforced through safe login protocols. Admins will have access to a dedicated User Management Dashboard where they can view all registered users, update profiles, reset forgotten passwords, deactivate accounts, or even delete users when necessary—ensuring full control over the system's access.

To maintain transparency and accountability, the system will track each user's login history and in-app activity, helping identify unusual behavior or misuse. If a user remains inactive for a long period, they will be automatically logged out to protect their account from unauthorized access.

For ease of use, the platform will include a secure password reset feature so users who forget their password can recover access without contacting support.

The admin dashboard will also display real-time insights on user activity, such as which roles are most active, which modules are being used, and system usage trends. These insights can help admins make decisions on training, support, or system improvement.

Over time, the system may also analyze feedback and behavior patterns to improve usability and performance. This ensures CocoSmart continuously evolves to meet the real needs of its users, whether they're managing plantation operations, handling deliveries, or placing product orders.

Feedback Management

In CocoSmart, the Feedback Management module plays an important role in building trust and improving customer satisfaction. After receiving their coconut-based products or experiencing delivery service, customers should be able to easily share their thoughts through the system. This includes leaving reviews, giving star ratings, or commenting on their overall experience. Whether it's a compliment, suggestion, or complaint, customers will have a simple, user-friendly form to submit their feedback directly through their account.

Each piece of feedback will be automatically linked to the customer and the specific order it relates to. This ensures that admins can trace feedback to actual events and take necessary actions. The system will organize all feedback neatly in an admin panel where feedback can be filtered by date, rating, category (e.g., complaint, suggestion), or even by product type.

To make sure customer concerns are addressed quickly, the system will send an automatic alert to the admin when negative feedback is received (e.g., low rating or a flagged complaint). Admins can reply directly through the platform, and customers will see these replies when they log into their account—making the process feel more personal and responsive.

CocoSmart will also generate customer satisfaction reports summarizing average ratings, common problems reported, areas that need improvement, and trends over time. These insights will help plantation owners or management teams refine their services and products. Feedback data can be exported for offline use, performance reviews, or meetings—ensuring the voice of the customer is always heard and valued.

Transport Coordination

The Transport Coordination module is designed to ensure that every order from the plantation reaches the customer efficiently and on time. Admins will have access to a dedicated dashboard where they can manage delivery operations—assigning transport jobs to available drivers or vehicles based on delivery location, urgency, and product type (e.g., fresh coconut, oil, fiber).

The system will intelligently group orders going to nearby destinations, optimizing delivery routes to save fuel and reduce travel time. This not only lowers transport costs but also helps in delivering fresher products to customers. A live delivery status system will track each order's progress and display clear updates like “Pending,” “Dispatched,” “On the Way,” and “Delivered.” These status updates will be visible to both the admin and the customer in real time.

All available drivers and vehicles will be listed in the system, showing their current availability and capacity. When a delivery is created, the system will calculate estimated transport cost based on the delivery distance, fuel consumption, and vehicle type. If there are any issues—such as unexpected delays, route disruptions, or missed deliveries—the system will instantly alert the admin, who can then adjust delivery plans as needed.

Additionally, the system will generate transport performance reports, showing metrics like the number of deliveries completed, percentage of on-time deliveries, total delivery costs, and driver-specific performance data. These reports will help in identifying bottlenecks and optimizing transport operations over time. Customers, too, will be able to track their delivery status directly from their dashboard, creating a more transparent and dependable experience.

Labor Management and Task Assignment

The Labor Management and Task Assignment function is a key part of the Smart Coconut Plantation Management System. Its goal is to manage the workforce effectively and ensure tasks are completed on time. The HR Manager leads this function, which focuses on digitizing manual labor processes. It manages labor profiles, assigns daily or scheduled tasks, monitors progress, and automates payroll generation. By improving coordination among human resources, this module plays a vital role in increasing productivity, transparency, and compliance on the coconut plantation.

The HR Manager can assign routine tasks to available workers. The system allows for immediate task assignment for the current day along with scheduling for future dates and times. It supports assigning tasks to multiple workers at once, and administrators can add notes or instructions related to each task. This feature helps allocate labor effectively, reduces miscommunication, and prevents delays in field operations. The task scheduling interface offers flexibility for managing recurring activities, rotating shifts, and temporary labor, reflecting the changing needs of plantation work. Once tasks are assigned, the system marks their status as “Pending” and records them in a central task database.

Workers can update the status of their tasks through the system interface, moving through stages like “Pending,” “In Progress,” and “Done.” This real-time interaction encourages accountability and allows supervisors to track task status without visiting each location. Notifications can be sent to workers when tasks are assigned or when status updates are overdue. Additionally, all task updates are time-stamped and logged, which helps with future audits and performance reviews.

A key feature of this module is automatic tracking of labor hours. The system logs the start and end times of tasks directly or based on worker updates. These recorded hours are used for payroll calculations. The system contains both hourly and daily wage settings and calculates total earnings for each worker for each pay cycle. By integrating payroll report generation, the system

reduces manual effort, minimizes human error, and ensures fair pay while maintaining transparency and compliance with labor laws.

This module can generate performance reports for each worker based on task completion, time efficiency, and consistency. These reports help the HR Manager identify high performers, pinpoint bottlenecks, or plan team rotations. The system also calculates labor cost reports using working hours and wage data, assisting management in making informed decisions about labor budgeting.

Sales and Finance Management

The Sales and Finance module of the CocoSmart system plays a central role in handling product sales and managing financial transactions. Its purpose is to simplify the buying experience for customers while also giving the plantation team full control over sales data and financial reporting.

At the heart of this module is the product management functionality. Authorized admins can add, update, or delete product listings, ensuring all key information such as product name, price, available quantity, image, and description is accurate and up to date. Products will be organized into categories, making it easier for customers to browse and filter based on what they're looking for.

Customers should be able to search for products, apply filters (like category or price range), and add items to a shopping cart. They should also be able to modify quantities or remove items before checking out. During checkout, users can choose their preferred payment method (either online or cash on delivery) and input their delivery details. Once an order is placed, the system automatically generates a digital invoice showing product details, and the final amount.

To keep customers informed, the module must also support real-time order tracking. Each order should go through a set of clearly defined statuses such as "Pending," "Packed," "Dispatched," and "Delivered" which are visible to both the customer and admin team. Staff members with proper access can update these statuses as the order progresses.

Promotions and seasonal discounts are also handled within this module. Admins should be able to define special offers by setting the percentage, duration, and conditions. These discounts should be automatically reflected during checkout, ensuring customers see the updated prices without confusion.

On the finance side, the system includes a sales dashboard for admins and financial analysts. This dashboard will display useful insights like total revenue, top-selling products, and overall financial trends. Each sale is automatically logged as income, while costs such as delivery, packaging, and marketing can be manually or automatically recorded. The system should

continuously calculate the financial balance, giving the team a clear, up-to-date view of the operation's financial health.

Moreover, financial data from other modules like Labor, Inventory, and Transport will feed into this one. This creates a single point of financial reporting, which helps management and stakeholders make better decisions based on consolidated data.

Lastly, the module will allow customers to view their past orders, making it easier to re-order frequent items. These records are securely stored and can only be accessed by the customer or authorized staff.

Non-functional Requirements

1. Performance

The system should respond to user actions (e.g., scheduling a task, loading dashboard) within 2–3 seconds. The chatbot should respond in 1 second for FAQ inquiries.

2. Scalability

The system should be capable of dealing with increasing data volumes (more trees, plots, tasks, users) without affecting performance. It should also be capable of managing multiple farms or plantations under one account if needed.

3. Availability

The system must have 99.9% uptime so that users can use it at any time, particularly during farm working hours. Cloud hosting and backup servers can be used to realize this.

4. Security

Data of users like farm information, worker details, and login details need to be protected. This encompasses role-based access control, encryption of passwords, and data transfer via HTTPS securely.

5. Usability

The system should be intuitive and simple to use for farmers with little technical knowledge. This includes a minimalistic UI and linear workflows.

6. Reliability

The system should run smoothly without crashing or malfunctioning, especially during critical tasks like scheduling or updates. There should be error-handling mechanisms.

7. Maintainability

The system should be easy to update and maintain, with modular code and proper documentation. This allows for the possibility of future enhancement, e.g., adding new crop types or chatbot features.

8. Data Integrity

Data of all types (task records, tree updates) should be valid and consistent, with validation to prevent duplication, data loss, or conflicting entries.

Technical Requirements

The Coconut Plantation Management System is built using a modern and scalable technology stack that supports responsive design, real-time operations, secure access, and efficient data handling. Below are the key technical requirements and technologies used:

1. Frontend

- Developed using React.js to create a fast, interactive, and responsive user interface.
- Optimized for mobile devices with dynamic forms and smooth input handling.

2. AI Chatbot

- Integrated AI chatbot to assist users with farming tips, system navigation, and general queries related to plantation operations.

3. Authentication

- Secure user login and role-based access using JWT (JSON Web Token) authentication.

4. Backend

- Node.js with Express.js handles server-side logic and business operations across all modules, including task assignment, order processing, and inventory control.
- Validations included to prevent conflicts like overlapping task times for workers.

5. Database

- MongoDB is used for storing structured data such as inventory items, workers, tasks, orders, and deliveries.
- Supports category-based storage, batch tracking, and foreign key constraints for worker assignments.

6. Inventory & QR Code Integration

- QR code generation for labeling inventory items.
- QR scanning via phone camera to quickly update or retrieve stock details.

7. Payments

- Multiple payment options integrated: Cash on Delivery, Credit/Debit Card, and PayPal.

8. Notifications

- Integrated with services like Twilio to send real-time email or SMS alerts for deliveries, order status, or low-stock warnings.

9. Reports & Analytics

- Dynamic reporting using Chart.js or similar libraries to visualize key metrics like stock movement, worker performance, and sales trends.

10. Order & Delivery Management

- RESTful APIs built for handling orders, drivers, and vehicles.
- UI includes dropdowns for vehicle and driver selection, and a calendar picker for delivery scheduling.

11. Hosting & Deployment

- Cloud-hosted using platforms like Render, Vercel, or AWS to ensure uptime, scalability, and easy deployment.

12. Version Control

- Managed using Git and GitHub for team collaboration and version tracking.

Literature review

For the literature review, we researched existing platforms like FarmERP, CropIn, and AgriApp. We compared our web application with others and gathered information about features we currently lack. We could not find any platform specifically designed for managing coconut plantations with dedicated roles such as HR Manager, Field Manager, and Inventory Manager.

In terms of inventory and sales management, most websites offer features to manage stock levels and sales. We found that FarmERP is the only platform that supports limited task assignment and partial payroll management. We did not find a function on CropIn or AgriApp for assigning tasks to individual workers or for updating task progress [10].

We also noticed that none of these platforms have a built-in marketplace for selling coconut products directly to customers. Our system has this feature, along with payment gateway integration. Additionally, only our system includes a delivery scheduling function for assigning vehicles and drivers.

Furthermore, we discovered that CropIn and AgriApp do not offer AI support for farming guidance or feedback management. FarmERP includes decision support but lacks an interactive chatbot. All the systems we reviewed allow basic weather-based planning, but our system features customizable weather API integration [11].

Function / Feature	FarmERP	CropIn	AgriApp	Our System
Labor Management	✓	✗	✗	✓
Task Assignment	✓	✗	✗	✓
Inventory Management	✓	✓	✓	✓
Sales Management	✓	✓	✓	✓
Built-in Marketplace	✗	✗	✗	✓
Transport Coordination	✗	✓	✗	✓
AI Chatbot Support	✗	✗	✗	✓
Feedback Management	✗	✓	✗	✓
Plantation Management	✓	✗	✗	✓

Table 1

Methodology

To ensure the successful delivery of the Coconut Plantation Management System (CocoSmart), we have adopted a structured software engineering methodology that covers the full development lifecycle starting from requirements gathering, through design and development, to testing and integration. This approach is well-suited for building flexible, scalable systems using the MERN stack (MongoDB, Express.js, React.js, and Node.js) [4].

We have chosen the Agile Software Development approach, specifically the Scrum framework, as our guiding methodology. Agile allows us to work in short, manageable development cycles, encouraging continuous collaboration with our client, regular feedback, and the ability to adapt to changing requirements. Agile practices will be applied throughout every phase of the project to ensure we stay responsive, efficient, and aligned with stakeholder expectations [1].

Agile Software Engineering Approach

We're following the Agile methodology, with a focus on the Scrum framework, as the core approach for our project. Agile supports us in delivering the system in small, manageable increments, which helps us stay flexible and continuously improve based on feedback. Since we're working closely with a real client whose needs may evolve over time, this approach is a great fit for CocoSmart. It allows our team to collaborate effectively across different roles while keeping the client involved throughout the development process.

Key Principles of Agile

- Customer collaboration over contract negotiation - Owners and managers can actively test early versions of our application and suggest improvements.
- Responding to change over following a plan - Be ready to add new features or change existing ones based on user needs.
- Working software over comprehensive documentation - Instead of spending too much time on planning, releasing small functional parts of the app quickly.
- Individuals and interactions over processes and tools - Our team can have frequent discussions instead of relying only on documentation or project management tools.

Agile Practices Applied in CocoSmart

- Sprint Planning - The development tasks will be split into manageable 2-week sprints, each targeting a set of prioritized deliverables.

- Daily Stand-ups - Team members will provide updates, report issues, and maintain alignment.
- Sprint Review and Retrospective - At the end of each sprint, progress will be demonstrated, and lessons learned will be discussed.
- Backlog Grooming - The product backlog will be updated and refined based on changing priorities or feedback from the client.

Why Agile for This Project ?

- CocoSmart involves five major modules, which benefit from Agile's ability to break work into smaller, iterative components.
- The system must adapt to real-time feedback from a real-world plantation client. Agile allows frequent validation of client requirements through sprint reviews and demos.
- Agile promotes early delivery of value, enabling deployment of high-priority modules like Marketplace or Plantation Management first, even before the full system is complete [2].

AGILE DEVELOPMENT PROCESS

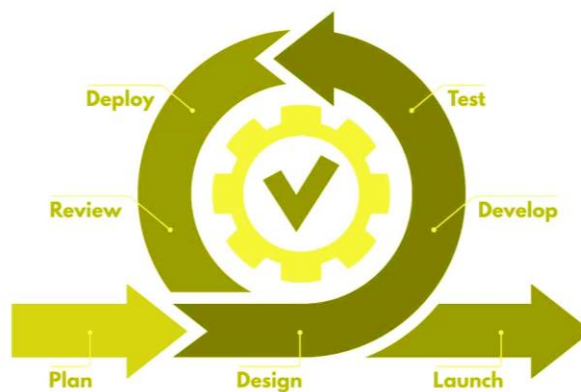


Figure 2

Requirements Engineering Methods

We will adopt Agile Requirement Gathering combined with User Stories and Stakeholder Interviews as our primary requirements engineering techniques. Agile allows for iterative development and continuous feedback from the client, which is ideal in a real-world project like this where requirements may evolve based on client needs [1], [2].

- User stories will help us understand what the users need by describing features from their point of view.
- Stakeholder Interviews and Observation sessions will provide context about existing workflows and challenges on-site.

Design Methods

We will use the Unified Modeling Language (UML) to represent the system's static and dynamic aspects. Diagrams such as Use Case, Flow charts, and ER Diagrams will be used to plan the structure and interactions within the system [3].

- Mockups (Figma) will be used to design user interfaces.
- We also plan to apply Component based Design principles in React, ensuring reusable UI components across the dashboard.

Development Tools and Technologies

For the development of CocoSmart, we have selected the MERN stack, which consists of MongoDB, Express.js, React.js and Node.js. This combination provides a comprehensive framework for building scalable and maintainable web applications.

- MongoDB - For NoSQL document-oriented database needs. It is suitable for storing varied data types and handling large sets of data [5].
- Express.js - A minimal and flexible Node.js web application framework that helps back-end development by providing essential tools for building APIs and web services efficiently [6].
- React.js - React will be used to create a responsive and interactive front end for our web application [7].
- Node.js - For the server environment and backend logic [8].

Other supporting tools include:

- Postman - Tool for testing and developing APIs. It allows users to send HTTP requests, view responses, and automate testing, making it easier to design, test, and debug APIs.
- VS Code - This versatile code editor will be used for writing and editing code across different programming languages [9].
- Git & GitHub - Version control will be handled using GitHub. GitHub provides a collaborative platform for code development and review.

The MERN stack is chosen due to its JavaScript consistency across all layers and fast development cycle [4].

Testing Methods

- Unit Testing - Testing individual components in isolation to ensure that each part of the application, such as the Payment & Return Handling module, functions correctly on its own.
- System Testing - Validating end-to-end system functionality to ensure that all components work together seamlessly, covering the integration of functionalities.
- User Acceptance Testing - Confirm that the system meets user requirements by having real users test the application and provide feedback, ensuring it meets the expectations of bookstore owners and customers.

Integration Methods

The system will be built using Modular Integration, where each module (e.g., Inventory, Marketplace, Plantation Management) is developed and tested independently and then integrated through API endpoints. RESTful APIs will ensure clean communication between the frontend and backend. We chose modular integration to reduce risk during development and allow parallel development efforts by team members.

Work Breakdown Structure (work distribution)

Major Task	Sub Task	Description
Requirements Engineering	Interviews, Surveys, Documentation	Gather and analyze stakeholder needs
System Design	Use case diagram, ER diagram, Wireframes	Plan the architecture and user interface
Frontend Development	Setup React, UI Components, Routing	Build user interface using React.js
Backend Development	API development, DB design, Authentication	Use Node.js + Express to connect frontend to MongoDB
Testing	Unit, Integration, UAT	Ensure modules work properly and meet requirements
Integration	Frontend and backend Integration	Connect frontend and backend with APIs
Finalization & Deployment	Hosting, Bug fixes, Documentation	Host app, submit reports

Table 2

Project Plan (Gantt Chart)

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Project Initiation and Planning								
Requirements Analysis								
Project Presentation								
System Design								
Development								
Testing								
Deployment								
Final Presentation and Viva								

Table 3

Our project, CocoSmart - Coconut Plantation Management System, is being developed following a carefully planned 8-week timeline to ensure that each stage of the system is delivered on time, with regular input from the client.

Week 1: Project Initiation and Planning

We kicked off the project by identifying the real-world challenges faced by coconut plantation owners especially issues like manual tracking, labor inefficiencies, and limited online sales. During this week, our team defined the project scope, goals, and deliverables. Early planning helped us stay focused and organized for the rest of the journey.

Week 2: Requirements Analysis

In this phase, we worked closely with our client to understand their expectations in detail. We conducted several discussions to gather and prioritize both functional and non-functional

requirements. By the end of the week, we had prepared a solid draft of the Software Requirements Specification (SRS) to be reviewed by the client.

Week 3: Project Presentation

We prepared and delivered our first formal project presentation, sharing everything we had worked on so far—from the problem statement and user requirements to our early design concepts. The feedback we received has helped shape our direction as we move into system design.

Weeks 4–5: System Design

In the upcoming weeks, we will focus on designing the full structure of the CocoSmart system. Our team will develop wireframes and mockups for the user interface, and design the database using ER/EER diagrams. We'll also define the backend and frontend architecture, keeping scalability and usability in mind. The finalized design will be reviewed with the client for approval.

Weeks 5–6: Development Phase

After the design is approved, we will begin coding the system. We plan to start with core modules such as plantation management, labor coordination, and inventory tracking. The frontend and backend will be developed in parallel, and we'll follow an agile approach with short development sprints to deliver working features regularly.

Weeks 6–7: Testing Phase

Once the core functionality is developed, we will thoroughly test the system. This will include both manual and automated testing to ensure everything works as expected. We'll identify and resolve bugs, improve performance, and make adjustments based on internal and client feedback.

Week 8: Deployment and Final Presentation

In the final week, we will deploy the complete CocoSmart system to a demo or live environment. We'll also prepare our final presentation and viva. During this session, we'll demonstrate the system's key features, explain its impact, and gather final feedback from our client and evaluators.

Evaluation Method

1. User Acceptance Testing (UAT)

Users from each role — admins, workers, processing staff, and customers — will test the system in real-life situations to check if it meets their needs. For example, admins will manage trees, workers will update their tasks, processing staff will log product batches and check inventory, and customers will place orders and give feedback. Their opinions will help make sure the system is useful and easy to use.

2. Functionality Testing

Each part of the system will be tested carefully to make sure everything works as it should:

- **Plantation & Tree:** Check if admins can add or change tree details, divide plantations into sections, and set reminders for fertilizer or pesticide. If there's a weather alert feature, test if alerts work correctly.
- **Labor & Task:** Make sure workers can register, log in, get tasks, update task status, and the system tracks work hours and creates reports correctly.
- **Processing & Inventory:** Confirm that staff can record processing details, set expiry dates, view stock levels, and get low-stock alerts.
- **Online Orders:** Test if customers can browse products, add to cart, place orders, track order status, and submit feedback.
- **User Management:** Check user registration and login, that admins can manage user permissions, and that the system tracks user activity properly.
- **Transport Coordination:** Verify that the system assigns deliveries, plans routes, tracks vehicles, and calculates transport costs correctly.

3. Performance Testing

Test how well the system works when many users use it at the same time or when handling lots of data:

- See if the system stays fast and responsive when workers update tasks, customers place orders, and inventory is managed all at once.
- Check if reports and alerts load quickly even with large amounts of data.

4. Usability Testing

Ask users with different computer skills to try the system. Watch if they can:

- Easily manage trees and tasks.
- Place orders and track deliveries without confusion.
- Use the feedback form properly.

If they find anything difficult, the design will be improved.

5. Security and Access Control Testing

Make sure only the right people can access certain parts of the system:

- Only admins can change plantation info, user data, and inventory.
- Workers can only see and update their own tasks.
- Customers can only place orders and view their own data.

Also, test password protection and data safety.

6. Feedback Collection and Improvement

After the system is live, collect feedback from users through forms and surveys. Use this feedback to fix problems, make the system easier to use, and add new features in updates.

7. Report Accuracy Check

Double-check that reports (like labor costs, stock levels, delivery status, and transport costs) show correct information by comparing them with manual calculations or sample data.

This evaluation plan helps ensure the system works well, is easy to use, secure, and meets the needs of everyone involved.

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Appendix

Figure 1 - System Overview Diagram

The System Overview Diagram gives a clear picture of how the CocoSmart platform works behind the scenes. It shows how different parts of the system like Plantation Management, Inventory, Marketplace, Labor Management, and User Services connect and support each other.

It also maps out how different users, such as farm managers, workers, buyers, and admins, interact with the platform, along with how CocoSmart links to external services like payments and notifications. This helps visualize how everything works together to make plantation operations smoother and more efficient.

Figure 2 - Agile Methodology

This figure illustrates the Agile methodology followed during the CocoSmart project. It emphasizes iterative development, where tasks are broken down into small sprints to ensure flexibility and continuous improvement. Regular feedback from stakeholders and sprint reviews helped the team adapt quickly and deliver a more user-focused system.

Table 1 - Literature Review Summary

This table shows the main sources we studied to understand existing systems. It helped us find useful ideas and identify what's missing, so we could improve our CocoSmart system.

Table 2 - Work Breakdown Structure

This table outlines the major phases of the CocoSmart project and breaks them down into smaller tasks. It shows what each stage involves from gathering requirements to deploying the final system. Each task plays a key role in building a fully functional, user-friendly platform for coconut plantation management.

Table 3 - Gantt Chart

This table presents a Gantt chart that outlines the timeline of the CocoSmart project over an 8-week period. It visually maps out when each major task will begin and end, helping the team stay on track and manage time effectively throughout the development process.